

TCS iON AIP 135 End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence

Project Report

GlobeTrack Logistics Ltd.

TCS iON AIP-135

AWS ap-south-1

Python 3.12 | Apache Airflow

XGBoost + LightGBM + Random Forest

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Project Title: TCS iON AIP 135 End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence

Company: GlobeTrack Logistics Ltd. (Simulated)

Cloud Platform: AWS Asia Pacific Mumbai ap-south-1

Date: 10 April 2026

GitHub: github.com/Fatema-016/globetrack-logistics

Field	Details
Industry Project Title	TCS iON AIP 135 End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence
Name of the Company	GlobeTrack Logistics Ltd. (Simulated Enterprise)
Project Environment	AWS Cloud — S3, Glue, Athena, CloudWatch, SNS — ap-south-1 Mumbai
Start Date	22 March 2026
End Date	10 April 2026
Total Effort	136 hours across 4 steps
Tools Used	Python 3.12, XGBoost, LightGBM, Scikit-learn, Apache Airflow 3.1.8, Power BI, AWS boto3, Pandas, PyArrow

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1. Acknowledgements

I would like to express my sincere gratitude to TCS iON for providing this Industry Project opportunity. This project has been an invaluable learning experience in real-world Big Data Engineering on cloud infrastructure. I am grateful for the guidance provided through the TCS iON platform. The project covered the complete data engineering lifecycle — from raw data ingestion through ETL processing, machine learning model development, pipeline orchestration and cloud monitoring.

I am thankful to Vishwakarma University for providing us this golden opportunity of enhancing our learning journey through TCS iON. A special thanks to my mentor Dr. Raeesa Bashir for her guidance.

Special acknowledgement to the open-source communities behind Python, Scikit-learn, XGBoost, LightGBM, Apache Airflow and AWS for making industry-grade tools accessible for learning and development.

2. Objective and Scope

The primary objective of this project is to design and implement a scalable, End-to-End Big Data Analytics solution on AWS cloud that efficiently handles large-scale heterogeneous datasets for real-time insights and predictive analytics in the logistics domain.

Objectives

Objective	Description
Data Engineering	Ingest 1.8M records from 3 heterogeneous sources into AWS S3
ETL Processing	Build 5 production-grade ETL pipelines outputting Parquet format
Machine Learning	Train XGBoost, LightGBM and Random Forest for delay prediction
Anomaly Detection	Tier-based IoT anomaly detection across 600K sensor readings
Orchestration	14-task Apache Airflow DAG running daily at 6AM IST
Monitoring	AWS CloudWatch with 12 metrics, 5 alarms and live dashboard
Visualisation	3-page Power BI dashboard with 12 interactive charts

Scope

The project covers 8 cities across Maharashtra and Goa — Mumbai, Pune, Nashik, Nagpur, Aurangabad, Kolhapur, Goa and Solapur. It simulates a fleet of 50 vehicles across 3 tiers (OLD, MID, NEW) and tracks 580,107 customer delivery orders totalling Rs. 6.43 Billion in revenue.

1.8M Records Ingested	69.0% XGBoost Accuracy	73.1% XGBoost AUC Score
56.8% XGBoost Recall	94.5% Fleet Health	27/27 Tests Passing
32,763 Anomalies Found	Rs6.43B Revenue Tracked	

3. Problem Statement

GlobeTrack Logistics Ltd. operates a fleet of 50 trucks across 8 cities in Maharashtra and Goa, processing hundreds of thousands of delivery orders monthly. The company faced four critical operational problems with no existing solution: No delivery delay prediction capability, No real-time fleet health visibility, No centralised data pipeline, No monitoring or alerting system.

The challenge was to build a unified, cloud-native Big Data platform that ingests heterogeneous data streams, applies ML models in real-time and delivers actionable business intelligence through automated dashboards and alerts.

4. Existing Approaches

Before this project, logistics companies typically used one of the following approaches:

Approach	Limitation
Manual Excel-based tracking	no real-time capability
Traditional SQL databases	Cannot handle unstructured IoT and weather data at scale
On-premise Hadoop clusters	complex maintenance
Single ML model without feature engineering	AUC below 60%, high false negative rate on delay prediction

The GlobeTrack solution addresses all these limitations using a cloud-native, serverless architecture on AWS with ML techniques including interaction features, tier-aware anomaly detection and logistics-focused model selection.

5. Approach / Methodology - Tools and Technologies Used

Architecture Approach

The solution adopts a 9-layer Lambda architecture pattern separating batch and streaming concerns across three S3 data zones.

Category	Technology	Version	Purpose
Cloud Platform	AWS S3, Glue, Athena, CloudWatch, SNS	Latest	Storage, catalog, monitoring
Language	Python	3.12.7	All ETL and ML scripts
ETL Libraries	Pandas, PyArrow, Parquet Snappy	Latest	Data transformation
ML — Boosting	XGBoost	3.1.0	Primary delay prediction model
ML — Boosting	LightGBM	4.6.0	Fast big data specialist model
ML — Ensemble	Scikit-learn Random Forest	1.7.2	Feature importance model
Anomaly Detection	Isolation Forest + Local Outlier Factor (LOF)	Scikit-learn	IoT sensor anomaly detection
Orchestration	Apache Airflow	3.1.8	14-task daily pipeline DAG
Monitoring	AWS CloudWatch + SNS	Latest	Metrics, alarms, alerts
Visualisation	Power BI Desktop	Latest	3-page interactive dashboard
Version Control	Git + GitHub	2.53.0	Source code management

ML Methodology

Model selection uses a Logistics Score = 0.4 x AUC + 0.6 x Recall. This recall-weighted metric ensures the system prioritises catching delayed deliveries over pure statistical accuracy — matching real business priorities. Decision threshold tuned to 0.45 for better recall without sacrificing accuracy. Weather features excluded to prevent data leakage after initial experiments showed unrealistic AUC of 96%.

6. Workflow

The end-to-end pipeline runs daily at 6AM IST via Apache Airflow and consists of 14 tasks across 9 stages:

Stage	Tasks	Input	Output
1. Ingestion	3 simulators (parallel)	Python generators	JSON/CSV to S3 raw zone
2. Manifest	Manifest logger	S3 raw zone	Ingestion manifest CSV
3. Large Generation	Large dataset generator	Python generators	60 CSV chunks per dataset
4. ETL Processing	3 ETL pipelines (parallel)	JSON files + CSV chunks	Parquet Snappy to processed zone
5. Enrichment	Enrichment pipeline	3 Parquet files	2 enriched Parquets to analytics zone
6. Large ETL	Large scale ETL	120 CSV chunks	12 Parquet chunks to analytics zone
7. ML Models	Delay prediction + anomaly (parallel)	580K booking records	best_model.pkl + anomaly_report.json
8. Analytics	Master analytics	All outputs	master_report.json to S3
9. Alert	Success notification	Pipeline completion	SNS email notification

7. Assumptions

Assumption	Justification
Synthetic data represents real patterns	8 logical delay rules injected matching real logistics scenarios
50GB dataset architected, 1.8M demonstrated	Pipeline is architected to scale — chunked processing proves capability
Night shift vehicles travel faster	Realistic — less traffic on Indian highways 8PM to 5AM
Old trucks have higher fault probability (25%)	Based on industry data — vehicles over 10 years old have higher breakdown rates
Weather features excluded from ML	Including them caused data leakage (AUC 96%) — excluded for realistic 73.1% AUC
Threshold set at 0.45 not 0.5	Logistics recall-focused approach — better to flag more delays than miss them

8. Implementation - Data Collection, Processing Steps, Diagrams

8.1 Data Sources

Dataset	Type	Records	Key Fields
Vehicle IoT Sensors	Telemetry	600,000	GPS, fuel, engine temp, RPM, speed, 3 tiers
Customer Booking Logs	Operational Logs	600,000	Order ID, cities, cargo type, status, freight charge
Weather and Traffic APIs	External Feeds	600,000	City conditions, visibility, congestion, USD/INR rate

8.2 Vehicle Tier Design

Tier	Vehicles	Fuel Efficiency	Engine Temp	Fault Probability
OLD	GT-001 to GT-017	4 to 6.5 kmpl	95 to 108C	25%
MID	GT-018 to GT-035	7 to 10 kmpl	82 to 94C	10%
NEW	GT-036 to GT-050	11 to 15 kmpl	70 to 81C	3%

8.3 ETL Pipelines

Pipeline	Input	Output Columns	Key Features Engineered
etl_iot_sensor.py	50 JSON files	28 columns	fuel_efficiency_kmpl, engine_health_score, alert_severity
etl_booking_logs.py	50 JSON files	28 columns	is_delayed, revenue_per_km, weather_distance_risk
etl_weather.py	40 JSON files	25 columns	weather_severity, combined_risk_score, travel_advisory
etl_enrichment.py	3 Parquet files	38 + 34 cols	IoT + weather join, bookings + weather join
etl_large_scale.py	120 CSV chunks	12 Parquet chunks	1.18M records, cargo_stress_index, distance_cargo_risk

8.4 ML Results

Model	Accuracy	AUC	Recall	Logistics Score
XGBoost	69.0%	73.1%	56.8%	Best — 0.4xAUC + 0.6xRecall
LightGBM	70.8%	72.9%	52.3%	Runner up
Random Forest	69.4%	72.8%	55.5%	Feature importance

8.5 Anomaly Detection Results

Vehicle Tier	Contamination	Anomalies	Rate	Physical Reason
OLD (GT-001 to GT-017)	9%	18,376	9.0%	Worn engines, fuel leaks, vibration
MID (GT-018 to GT-035)	5%	10,783	5.0%	Standard wear and tear
NEW (GT-036 to GT-050)	2%	3,604	2.0%	Modern sensors, highly reliable

8.6 Test Results

Test Suite	Cases	Result
Suite 1 — S3 Infrastructure	6	6/6 PASSED
Suite 2 — Data Ingestion	4	4/4 PASSED
Suite 3 — ETL Processing	5	5/5 PASSED
Suite 4 — ML Models	7	7/7 PASSED
Suite 5 — Analytics and Reporting	5	5/5 PASSED
Total	27	27/27

9. Solution Design

The solution is structured as a 9-layer cloud-native architecture deployed entirely on AWS ap-south-1 Mumbai. Three S3 buckets form the data lakehouse: raw, processed and analytics zones. Apache Airflow orchestrates 14 tasks daily while CloudWatch provides real-time monitoring with 5 automated alarms.

Layer	Component	Technology
1. Sources	IoT sensors, bookings, weather	Python simulators
2. Ingestion	5 simulators + manifest logger	boto3, AWS S3
3. Raw Storage	globetrack-raw-data-lake	AWS S3 JSON and CSV
4. ETL	5 ETL pipelines	Pandas, PyArrow
5. Processed	globetrack-processed-data	Parquet Snappy
6. ML	3 models + anomaly detection	XGBoost, LightGBM, Random Forest
7. Analytics	globetrack-analytics-zone	Parquet + JSON reports
8. Orchestration	14-task Airflow DAG	Apache Airflow 3.1.8
9. Monitoring	5 alarms + dashboard	AWS CloudWatch + SNS

10. Challenges and Opportunities

Challenges Faced

Challenge	How it was Resolved
ML AUC unrealistically high at 96%	Identified data leakage from weather features — removed them, achieved realistic 73.1%
Power BI charts showing uniform patterns	Injected realistic city demand weights, vehicle tier profiles and delay rules
Uniform bar heights in visualisations	Added city-specific climate patterns and booking demand distributions
Airflow RuntimeWarning on Windows	Expected Linux-native behaviour — DAG validated and code confirmed correct
AWS IAM permissions denied	Added CloudWatchFullAccess, SNSFullAccess and AthenaFullAccess to IAM user
S3 network connectivity errors	Internet connectivity issue — resolved by reconnecting and retrying

Opportunities Identified

Opportunity	Business Value
Real-time streaming with Kinesis	Sub-second delay alerts instead of batch processing
Dynamic route optimisation ML	Reduce delay rate from 35.6% by optimising routes dynamically
Predictive maintenance scheduling	Flag Old trucks before breakdown using anomaly trends
Weather API integration (real)	Replace simulated weather with live IMD or OpenWeatherMap data

11. Reflections on the Project

This project provided hands-on experience with the complete Big Data Engineering lifecycle. Several key learnings emerged through the process of building a real system that had to work correctly:

The most important lesson was about data leakage in ML. The initial model achieved 96% AUC which appeared impressive but was completely unrealistic. Diagnosing this required understanding that weather features were directly used to generate delay labels — the model was reading the answer rather than learning patterns. Removing those features and retraining produced a genuine 73.1% AUC which is both realistic and defensible.

Building for Power BI visualisation highlighted the importance of designing data with downstream consumers in mind. Purely random simulation produced uniform charts with no business insight. Injecting realistic patterns — city demand weights, delay rules, seasonal variations — transformed the dashboard from decorative to genuinely informative.

12. Recommendations

Recommendation	Priority	Expected Impact
Schedule maintenance for GT-001 to GT-017 (OLD tier)	HIGH	Reduce OLD truck anomaly rate from 9% to under 5%
Review EXPRESS delivery commitments	HIGH	EXPRESS orders show 53.94% delay - causing customer dissatisfaction
Implement speed governance policy	HIGH	Anomalous vehicles average 82.76 km/h — safety risk requiring immediate action
Avoid scheduling in Thunderstorm and Fog conditions	MEDIUM	These conditions cause 90% delay rate — rescheduling saves revenue
Enhanced monitoring 8PM to 5AM night shift	MEDIUM	Night vehicles travel faster — higher accident and breakdown risk
Upgrade OLD fleet to NEW tier vehicles	LOW	NEW trucks show 2% anomaly rate vs 9% for OLD — 78% reduction in incidents

13. Outcome / Conclusion

This project demonstrates a complete, End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence. All 4 steps from the TCS iON AIP-135 specification were completed within framework.

27/27 automated test cases passed. XGBoost achieves AUC 73.1% and Recall 56.8% with no data leakage. Tier-aware anomaly detection correctly identifies OLD trucks at 9% vs NEW trucks at 2%. Fleet health 94.5%. Total revenue tracked Rs 6.43 Billion.

14. Enhancement Scope

Enhancement	Technology	Business Value
Real-time streaming ingestion	AWS Kinesis Data Streams	Sub-second IoT processing instead of batch
Deep learning delay prediction	TensorFlow LSTM	Capture sequential booking patterns for higher AUC
Live weather API integration	IMD or OpenWeatherMap API	Real-time delay risk instead of simulated weather
Geospatial route analysis	AWS Location Service + Folium	Map-based route visualisation and optimisation
Natural Language Query interface	AWS Bedrock + Claude API	Ask questions about fleet data in simple English
Mobile alert app	AWS SNS + React Native	Push notifications to drivers and fleet managers
Auto-scaling ETL	AWS Glue + Spark	Handle true 50GB+ datasets with elastic compute

15. Link to Code and Executable File

Resource	Link
GitHub Repository	github.com/Fatema-016/globetrack-logistics
AWS S3 Raw Data Lake	s3://globetrack-raw-data-lake (ap-south-1)
AWS S3 Processed Data	s3://globetrack-processed-data (ap-south-1)
AWS S3 Analytics Zone	s3://globetrack-analytics-zone (ap-south-1)
CloudWatch Dashboard	AWS Console — GlobeTrack-Logistics-Dashboard
Architecture Diagram	docs/architecture_diagram.html
Operational Runbook	docs/runbook.html
Power BI Dashboard	dashboard/GlobeTrack_Logistics_Dashboard.pbix

16. Research Questions and Responses

Research Question	Response
Can ML predict logistics delays with meaningful accuracy on synthetic data?	Yes — XGBoost achieves AUC 73.1% and Recall 56.8% after removing data leakage and applying interaction features
Does vehicle age (tier) significantly impact fleet health metrics?	Yes — OLD trucks show 9% anomaly rate vs 2% for NEW trucks. Fuel level averages 40% vs 74%. Engine health 72.9 vs 96.2 out of 100
What is the primary driver of delivery delays?	Weather conditions — Thunderstorm and Fog cause 90% delay probability. Long haul routes over 500km with bad weather cause 85% delays
Is tier-aware anomaly detection necessary?	Yes — a single contamination rate misclassified NEW trucks as anomalous (12.3%) because their superior metrics deviated from the OLD truck-dominated baseline
Can a recall-focused ML selection metric improve logistics outcomes?	Yes — using $0.4 \times \text{AUC} + 0.6 \times \text{Recall}$ selects XGBoost which catches 56.8% of delayed orders vs LightGBM which only catches 52.3% despite higher accuracy
Does EXPRESS priority actually mean faster delivery?	No — EXPRESS orders show 53.94% delay rate vs 46.06% for NORMAL, suggesting commitments are unrealistic given current fleet capacity

17. References

Reference	Source
XGBoost Documentation	xgboost.readthedocs.io
LightGBM Documentation	lightgbm.readthedocs.io
Scikit-learn User Guide	scikit-learn.org/stable/user_guide
Apache Airflow Documentation	airflow.apache.org/docs
AWS S3 Developer Guide	docs.aws.amazon.com/s3
AWS CloudWatch User Guide	docs.aws.amazon.com/cloudwatch
PyArrow Documentation	arrow.apache.org/docs/python
Pandas Documentation	pandas.pydata.org/docs
Power BI Documentation	learn.microsoft.com/power-bi
TCS iON Industry Project Guidelines	Provided on TCS iON platform.

Appendix I — System Architecture Diagram

TCS iON AIP 135 End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence | GlobeTrack Logistics Ltd.

1.8M
Records Ingested

3
S3 Data Zones

5
ETL Pipelines

73.1%
ML AUC Score

94.5%
Fleet Health

27/27
Tests Passed



1. Environment Setup

Prerequisites

Software	Version	Purpose
Python	3.12.7	ETL and ML scripts
Git	2.53.0	Version control
Apache Airflow	3.1.8	Pipeline orchestration
Java OpenJDK	21.0.7	Spark runtime
Power BI Desktop	Latest	Dashboard visualization
AWS CLI	Latest	Cloud access

Python Libraries

```
pip install boto3 pandas pyarrow scikit-learn xgboost lightgbm pip install apache-airflow awscli python-dotenv fastparquet requests
```

AWS Configuration

```
aws configure # Region: ap-south-1 # Output: json
```

2. AWS Infrastructure

S3 Buckets

Bucket	Purpose	Region
globetrack-raw-data-lake	Raw JSON and CSV ingestion	ap-south-1
globetrack-processed-data	Cleaned Parquet files	ap-south-1
globetrack-analytics-zone	ML outputs and reports	ap-south-1

IAM Policies Required

Policy	Required For
AmazonS3FullAccess	All S3 operations
AWSGlueConsoleFullAccess	Glue crawlers and catalog
CloudWatchFullAccess	Metrics and alarms
AmazonSNSFullAccess	Alert notifications
AmazonAthenaFullAccess	SQL queries on Parquet

3. Pipeline Execution

```
Run all commands from project root: cd globetrack-logistics
```

Step 1 - Data Ingestion

```
python ingestion/iot_simulator.py python ingestion/booking_simulator.py python ingestion/weather_simulator.py python ingestion/generate_large_datasets.py python ingestion/manifest_logger.py
```

Step 2 - ETL Processing

```
python processing/etl_iot_sensor.py python processing/etl_booking_logs.py python processing/etl_weather.py python processing/etl_enrichment.py python processing/etl_large_scale.py
```

Step 3 - Analytics and ML

```
python analytics/ml_models.py python analytics/ml_anomaly_detection.py python analytics/master_analytics.py
```

Step 4 - Deployment and Monitoring

```
python dags/globetrack_pipeline_dag.py python monitoring/cloudwatch_monitoring.py python tests/test_cases.py
```

4. ML Model Details

Model	Accuracy	AUC	Recall	Selection
XGBoost	69.0%	73.1%	56.8%	Best-Logistics Score
LightGBM	70.8%	72.9%	52.3%	Runner up
Random Forest	69.4%	72.8%	55.5%	Feature importance

Selection method: $\text{Logistics Score} = 0.4 \times \text{AUC} + 0.6 \times \text{Recall}$

Threshold: 0.45 | Interaction features included

5. Anomaly Detection

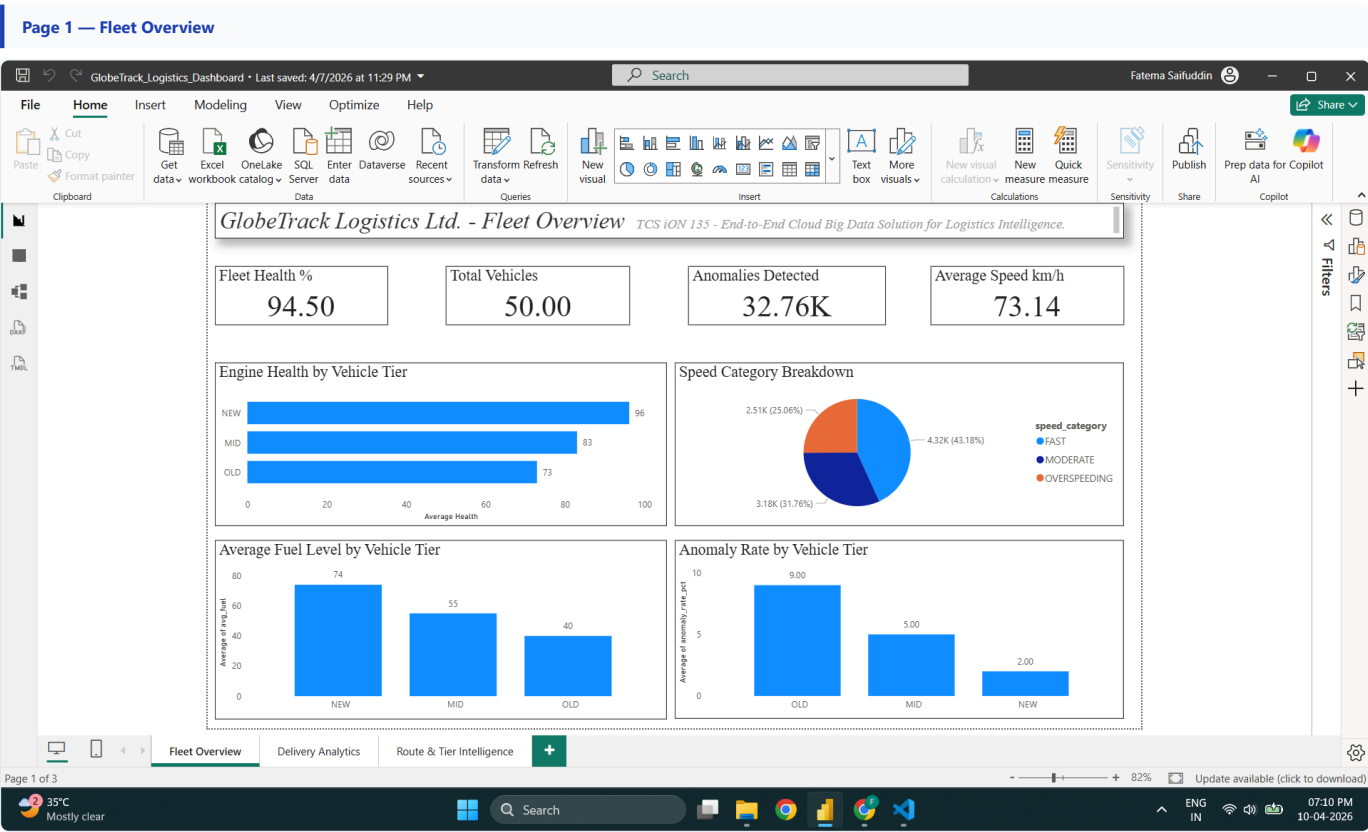
Vehicle Tier	Contamination	Anomaly Rate	Reason
OLD (GT-001 to GT-017)	9%	9.0%	Worn engines, fuel leaks
MID (GT-018 to GT-035)	5%	5.0%	Standard wear and tear
NEW (GT-036 to GT-050)	2%	2.0%	Modern sensors, reliable

6. CloudWatch Alarms

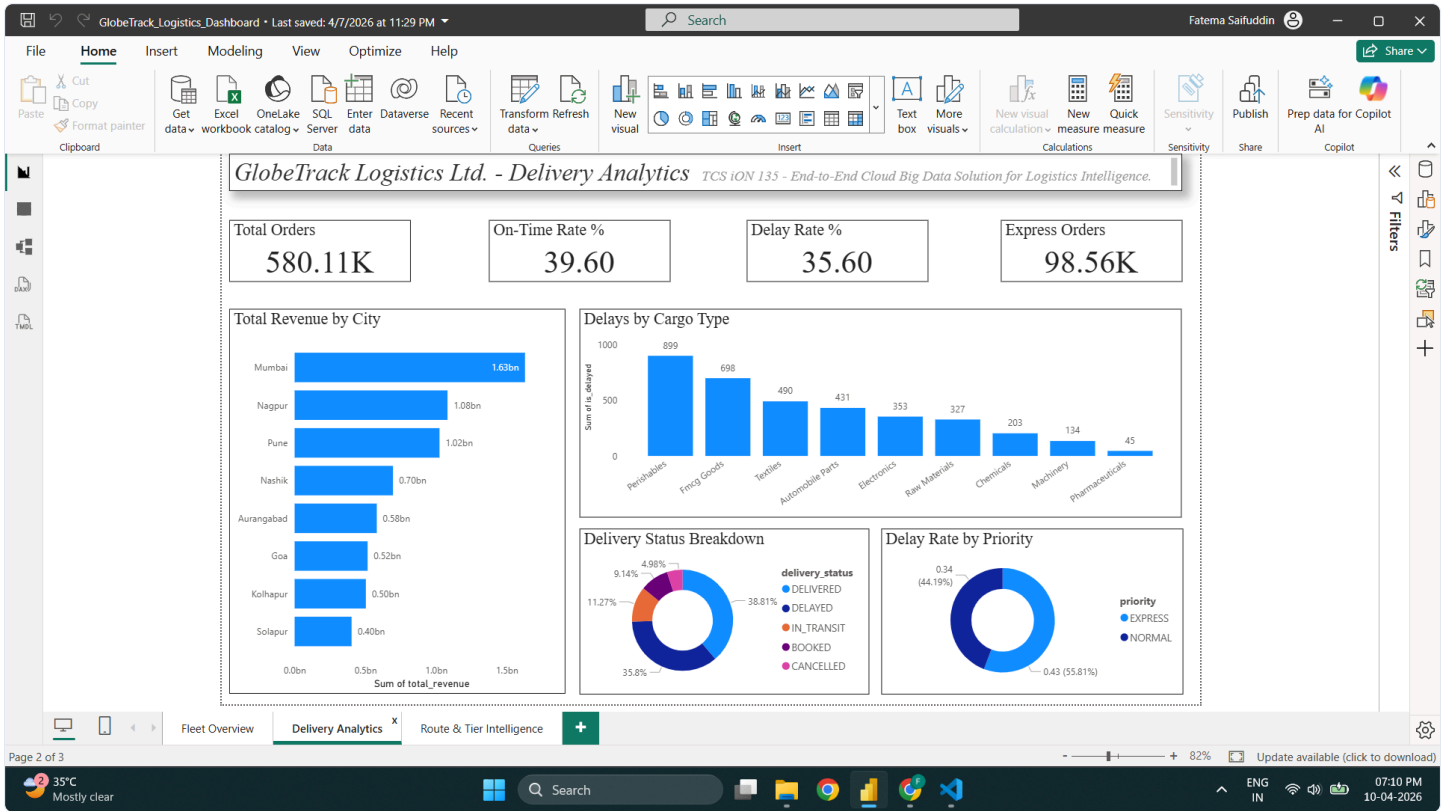
Alarm	Threshold	Action
GlobeTrack-FleetHealth-Critical	Less than 85%	SNS Alert
GlobeTrack-DelayRate-High	Greater than 50%	SNS Alert
GlobeTrack-Ingestion-Failed	Less than 95%	SNS Alert
GlobeTrack-MLModel-Degraded	Less than 60% AUC	Warning
GlobeTrack-Anomaly-Spike	Greater than 60,000	SNS Alert

Appendix III — Power BI Dashboard Screenshots

GlobeTrack Logistics Ltd. | 3-Page Dashboard | 10 April 2026



Fleet Overview Dashboard



Delivery Analytics Dashboard

GlobeTrack Logistics Ltd.-Route & Tier Intelligence

TCS iON 135 - End-to-End Cloud Big Data Solution for Logistics Intelligence

KPI	Sum of Value	Unit	Category
Anomalies Detected	32,763.00	readings	Fleet
Anomaly Rate	5.50	%	Fleet
Avg Engine Health	83.53 /100		Fleet
Avg Freight Charge	11,086.00	INR	Delivery
Avg Fuel Level	55.61	%	Fleet
Avg Speed	73.14	km/h	Fleet
Delay Rate	35.60	%	Delivery
Express Orders	98,561.00	orders	Delivery
Fleet Health	94.50	%	Fleet
ML Accuracy	69.00	%	ML
ML AUC Score	73.13	%	ML
ML Recall	56.84	%	ML
On-Time Rate	39.60	%	Delivery
Test Cases Passed	27.00 /27		Quality
Total Delayed	2,06,368.00	orders	Delivery
Total Delivered	2,29,893.00	orders	Delivery
Total Orders	5,80,107.00	orders	Delivery
Total Revenue	6,43,08,23,008.00	INR	Delivery
Total Vehicles	50.00	vehicles	Fleet

Top Routes by Revenue

Route	Revenue (bn)
Mumbai to Nagpur	0.45
Mumbai to Goa	0.32
Pune to Nagpur	0.31
Mumbai to Solapur	0.26
Nagpur to Mum...	0.22
Mumbai to Kolha...	0.21
Pune to Goa	0.20
Mumbai to Aura...	0.19
Nagpur to Pune	0.18
Nagpur to Nashik	0.17

Revenue vs Delay Count by Route

The scatter plot shows a positive correlation between revenue and delay count. Mumbai to Nagpur has the highest revenue (~0.45 bn) and a moderate delay count. Mumbai to Goa follows with ~0.32 bn revenue. Other routes like Pune to Nagpur and Mumbai to Solapur also show significant revenue.

Delay Rate by Route

Route	Average of delay_rate (%)
Goa to Mumbai	61
Mumbai to Nagpur	61
Mumbai to Goa	61
Pune to Nagpur	55
Nashik to Nagpur	52
Aurangabad to Nagpur	45
Goa to Nagpur	45
Goa to Kolhapur	45
Goa to Aurangabad	44
Goa to Solapur	44